

Heteroskedasticity

Intermediate Econometrics

Chengye Jia

Department of Economics
Shandong University of Finance and Economics

Spring 2026

Wooldridge, Chapter 8

Outline

- Motivation and Definition (§8-0)
- Consequences for OLS (§8-1)
- Heteroskedasticity-Robust Inference (§8-2)
- Testing for Heteroskedasticity (§8-3)
- Weighted Least Squares (§8-4)
- The Linear Probability Model Revisited (§8-5)
- Summary and Decision Framework

Motivation and Definition (§8-0)

What Is Heteroskedasticity?

Recall **MLR.5 (Homoskedasticity)**:

$$\text{Var}(u \mid x_1, \dots, x_k) = \sigma^2 \quad (\text{constant for all observations})$$

Heteroskedasticity

MLR.5 is **violated** when the error variance depends on the regressors:

$$\text{Var}(u_i \mid x_{i1}, \dots, x_{ik}) = \sigma_i^2$$

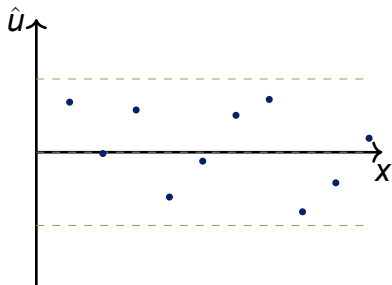
where σ_i^2 **varies** across observations $i = 1, \dots, n$.

Why does this happen?

- ▶ Richer households have *more variable* consumption (income effect on spread)

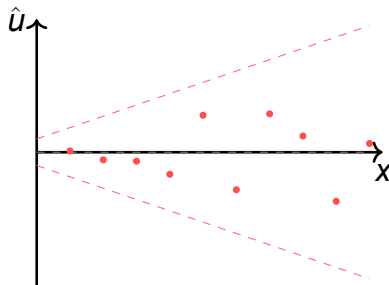
Visualising Heteroskedasticity

Homoskedastic errors



Constant band

Heteroskedastic errors



Fan-shaped spread

Residual plots ($\hat{u}_i$ vs. $\hat{y}_i$ or $\hat{u}_i^2$ vs. x_i) are the first diagnostic. A funnel or fan shape signals heteroskedasticity.

Consequences for OLS (§8-1)

Consequences: What Changes, What Stays the Same

NOT affected:

- ▶ **Unbiasedness** of $\hat{\beta}_j$ (only MLR.1–4 needed)
- ▶ **Consistency** of $\hat{\beta}_j$
- ▶ Interpretation of R^2 and $\bar{R}^2$

⇒ OLS point estimates are *still useful*.

IS affected:

- ▶ OLS **standard errors** are wrong
- ▶ ***t*-tests** and ***F*-tests** are invalid
- ▶ OLS is **no longer BLUE** (Gauss-Markov fails)
- ▶ **Efficiency** is lost

⇒ Inference based on usual SEs is *unreliable*.

Why OLS Standard Errors Are Wrong: The Proof

True variance of $\hat{\beta}_1$ in simple regression (derived from $\hat{\beta}_1 = \beta_1 + \sum_i w_i u_i$):

$$\text{Var}(\hat{\beta}_1) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2 \sigma_i^2}{\text{SST}_x^2} \quad (\text{eq. 8.2})$$

where $\text{SST}_x = \sum_i (x_i - \bar{x})^2$.

Under homoskedasticity ($\sigma_i^2 = \sigma^2$ for all i), this simplifies to:

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2 \sum_i (x_i - \bar{x})^2}{\text{SST}_x^2} = \frac{\sigma^2}{\text{SST}_x}$$

OLS uses $\hat{\sigma}^2 = \text{SSR}/(n - k - 1)$ to estimate σ^2 . If $\sigma_i^2 \neq \sigma^2$, then $\hat{\sigma}^2$ estimates a **weighted average** of σ_i^2 , not the right quantity for eq. 8.2.

Direction of bias: *unpredictable* without knowing how σ_i^2 relates to leverage $(x_i - \bar{x})^2$.

Gauss-Markov Fails: OLS Is No Longer BLUE

Recall the Gauss-Markov Theorem requires MLR.1–5. Under heteroskedasticity:

There exist **other linear unbiased estimators** with *smaller variance* than OLS.

Specifically, **Weighted Least Squares** (WLS) with the correct weights achieves lower variance than OLS.

Intuition:

- ▶ Observations with *small* σ_i^2 are very informative — OLS under-uses them.
- ▶ Observations with *large* σ_i^2 are noisy — OLS over-uses them.
- ▶ WLS corrects this by weighting inversely proportional to variance.

Even though OLS estimates remain unbiased, an analyst using OLS un-

Heteroskedasticity-Robust Inference (§8-2)

The White Robust Variance Estimator

Key insight (White, 1980): Replace the unknown σ_i^2 in eq. 8.2 with $\hat{u}_i^2$ (the squared OLS residual for observation i).

Heteroskedasticity-Robust Variance Estimator (eq. 8.4)

$$\widehat{\text{Var}}_{\text{rob}}(\hat{\beta}_j) = \frac{\sum_{i=1}^n \hat{r}_{ij}^2 \hat{u}_i^2}{\text{SSR}_j^2}$$

- ▶ $\hat{r}_{ij}$ = residuals from regressing x_j on *all other regressors*
- ▶ $\text{SSR}_j = \sum_i \hat{r}_{ij}^2$
- ▶ $\hat{u}_i$ = OLS residuals from the main regression

▶ Also called **White SE**, **Hetero-White SE**, **HQ standard errors**

Why the Robust Estimator Is Consistent: Sketch

We need: A consistent estimator of $\sum_i \hat{r}_{ij}^2 \sigma_i^2 / \text{SSR}_j^2$.

Step 1. Note that $\text{plim } n^{-1} \text{SSR}_j = \text{Var}(r_j)$ (a constant), so SSR_j^2 grows at rate n^2 .

Step 2. For each i : $\hat{u}_i = u_i + (\beta - \hat{\beta})'x_i$. Under MLR.1–4 and consistency of $\hat{\beta}$:

$$\hat{u}_i^2 \stackrel{p}{\approx} u_i^2 \quad \text{as } n \rightarrow \infty$$

Step 3. Therefore:

$$\frac{1}{n} \sum_i \hat{r}_{ij}^2 \hat{u}_i^2 \xrightarrow{p} \mathbb{E}[\hat{r}_{ij}^2 u_i^2] = \mathbb{E}[\hat{r}_{ij}^2 \sigma_i^2]$$

The robust estimator substitutes $\hat{u}_i^2$ for σ_i^2 term-by-term. By the Law of Large Numbers, the average converges to the correct quantity. No assumption about the *form* of σ_i^2 is required.

Robust t and F Statistics

Once we have robust SEs, all inference proceeds with the usual formulas.

Robust t -statistic:

$$t_{\text{rob}} = \frac{\hat{\beta}_j - \beta_j^0}{\widehat{\text{se}}_{\text{rob}}(\hat{\beta}_j)} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{under } H_0, n \rightarrow \infty$$

Same critical values as the standard t -test — only the SE changes.

Robust F -statistic (Wald form) — for q restrictions $H_0 : \mathbf{R}\beta = \mathbf{r}$:

$$F_{\text{rob}} = \frac{1}{q} (\mathbf{R}\hat{\beta} - \mathbf{r})' (\mathbf{R} \hat{V}_{\text{rob}} \mathbf{R}')^{-1} (\mathbf{R}\hat{\beta} - \mathbf{r}) \xrightarrow{d} F_{q, \infty} \quad \text{under } H_0$$

$\hat{V}_{\text{rob}}$: robust variance-covariance matrix using White estimator block.

Example 8.1: OLS vs. Robust SEs — Log Wage Equation

Dataset: WAGE1 ($n = 526$, $R^2 = 0.461$). Compare OLS SE () and robust SE [].

Log Wage Regression

Variable	Coef.	OLS SE	[Robust SE]
<i>marrmale</i>	0.213	(0.055)	[0.057]
<i>marrfem</i>	-0.198	(0.058)	[0.058]
<i>singfem</i>	-0.110	(0.056)	[0.057]
<i>educ</i>	0.0789	(0.0067)	[0.0074]
<i>exper</i>	0.0268	(0.0052)	[0.0051]
<i>exper</i> ²	-0.00054	(0.00011)	[0.00011]
<i>tenure</i>	0.0291	(0.0068)	[0.0069]
<i>tenure</i> ²	-0.00053	(0.00023)	[0.00024]
intercept	0.321	(0.100)	[0.109]

Example 8.1: Interpretation

- ▶ Robust SE for *educ*: $0.0067 \rightarrow 0.0074$ (slightly larger — mild heteroskedasticity present).
- ▶ Most SEs are nearly identical — significance conclusions unchanged.
- ▶ *marrmale* premium: 21.3% wage gap; *marrfem* penalty: -19.8% .
- ▶ Lesson: always report robust SEs and verify they do not alter inference.

When OLS and robust SEs are close, heteroskedasticity is **mild**. When they diverge substantially, inference from usual SEs is unreliable — robust SEs are preferred.

Example 8.2: Robust F -Test for Race Dummies

Dataset: GPA3 ($n = 366$). Dep. var.: *cumgpa*.

Cumulative GPA Regression

Variable	Coef.	OLS SE	[Robust SE]
<i>sat</i>	0.00114	(0.00018)	[0.00019]
<i>hsperc</i>	-0.00857	(0.00124)	[0.00140]
<i>tothrs</i>	0.00250	(0.00073)	[0.00073]
<i>female</i>	0.303	(0.059)	[0.059]
<i>black</i>	-0.128	(0.147)	[0.118]
<i>white</i>	-0.059	(0.141)	[0.110]
intercept	1.47	(0.23)	[0.22]
$R^2 = 0.4006$			

Testing $H_0 : \beta_{black} = \beta_{white} = 0$: Usual $F = 0.40$ ($p = 0.672$); Robust $F \approx 0.75$ — both fail to reject. Robust SEs for race dummies are *smaller*.

Heteroskedasticity-Robust LM Test (§8-2a)

LM tests can be made robust without matrix algebra.

Robust LM Test Procedure (for testing q exclusion restrictions):

1. Estimate the **restricted** model; save residuals $\tilde{u}_i$.
2. Regress each **excluded** variable on all **included** variables; save residuals $\tilde{r}_{i\ell}$ for $\ell = 1, \dots, q$.
3. Regress **1** on $\tilde{r}_{i1}\tilde{u}_i, \dots, \tilde{r}_{iq}\tilde{u}_i$ (no intercept); get SSR_0 .
4. Robust LM = $n - \text{SSR}_0 \xrightarrow{d} \chi_q^2$ under H_0 .

Key insight: the procedure uses products $\tilde{r}_{i\ell}\tilde{u}_i$ to “orthogonalise” the score. This makes the statistic valid under any form of heteroskedasticity — no variance model is assumed.

Example 8.3: Robust LM vs. Usual LM (CRIME1)

Dataset: CRIME1 ($n = 2725$). Test $H_0 : \beta_{avg_{sen}} = \beta_{avg_{sen}^2} = 0$.

Arrest Rate Equation — LM Test Comparison

- ▶ Usual $LM = 3.52$ ($p \approx 0.17$) — **fail to reject** H_0
- ▶ Robust $LM = 5.10$ ($p \approx 0.078$) — **borderline reject** at 10%

Heteroskedasticity *distorts* the usual LM test. The robust version detects evidence against H_0 that the usual test misses. Using the wrong test can lead to incorrect conclusions about which variables belong in the model.

Testing for Heteroskedasticity (§8-3)

Should We Always Use Robust SEs?

Case for always-robust:

- ▶ Valid under both homo- and heteroskedasticity
- ▶ No cost to unbiasedness
- ▶ Standard in modern applied work (cross-section data)

Case for testing first:

- ▶ Robust SEs can be imprecise in small samples
- ▶ Homoskedasticity $\Rightarrow$ OLS t -test is exact in finite samples (MLR.1–6)
- ▶ Knowing the variance form enables efficient WLS

Wooldridge's recommendation:

1. **Always** report robust SEs in empirical work with cross-sectional data.
2. Run a formal test if you want to understand the *form* of

The Breusch-Pagan Test: Setup

Null hypothesis: $H_0 : \text{Var}(u \mid x_1, \dots, x_k) = \sigma^2$ (homoskedasticity)

Idea: Under H_0 , u_i^2 should be unrelated to $x_1, \dots, x_k$. Test this with a regression.

BP Test Auxiliary Regression (eq. 8.12)

$$\hat{u}_i^2 = \delta_0 + \delta_1 x_{i1} + \delta_2 x_{i2} + \dots + \delta_k x_{ik} + v_i$$

Joint null: $H_0 : \delta_1 = \delta_2 = \dots = \delta_k = 0$ (eq. 8.13)

Why squared residuals? $\hat{u}_i^2$ is the sample proxy for $\sigma_i^2 = \text{Var}(u_i \mid x_i)$. If the latter depends on x , so should $\hat{u}_i^2$.

The Breusch-Pagan Test: Test Statistics

Two equivalent test statistics:

F-statistic from the auxiliary regression [8.12]:

$$F = \frac{R_{\hat{u}^2}^2/k}{(1 - R_{\hat{u}^2}^2)/(n - k - 1)} \sim F_{k, n-k-1} \quad \text{under } H_0$$

LM statistic (Lagrange Multiplier):

$$LM = n \cdot R_{\hat{u}^2}^2 \xrightarrow{d} \chi_k^2 \quad \text{under } H_0 \quad (\text{eq. 8.16})$$

where $R_{\hat{u}^2}^2$ is the R^2 from regressing $\hat{u}_i^2$ on $x_1, \dots, x_k$.

Limitation: BP detects only *linear* dependence of σ^2 on x .

Example 8.4: BP Test — Levels vs. Logs

Level Model (HPRICE1)

$$\widehat{price} = -21.77 + 0.00207 \text{ lotsize} \\ + 0.123 \text{ sqrft} + 13.85 \text{ bdrms}$$

$$n = 88, R^2 = 0.672$$

$$\text{BP } F = 5.34, p = 0.002$$

$$\text{BP } LM = 14.09, p = 0.003$$

Reject H_0 — heteroskedastic.

Log-Log Model (HPRICE1)

$$\log(\widehat{price}) = -1.30 + 0.168 \log(\text{lotsize}) \\ + 0.700 \log(\text{sqrft}) + 0.037 \log(\text{bdrms})$$

$$n = 88, R^2 = 0.643$$

$$\text{BP } F = 1.41, p = 0.245$$

$$\text{BP } LM = 4.22, p = 0.239$$

Fail to reject H_0 .

Log-transforming the dependent variable *stabilises the variance*. Level

The White Test: Motivation

Problem with BP: Misses heteroskedasticity that depends on *squares and cross-products* of regressors.

White (1980) proposed a more general auxiliary regression:

White Test — Full Form (eq. 8.19)

With regressors $x_1, x_2, \dots, x_k$, regress $\hat{u}_i^2$ on all unique terms:

$$x_1, x_2, \dots, x_k, \quad x_1^2, x_2^2, \dots, x_k^2, \quad x_1x_2, x_1x_3, \dots, x_{k-1}x_k$$

Joint F -test or $LM = nR_{\hat{u}^2}^2 \xrightarrow{d} \chi_m^2$ where m = number of slope terms.

Cost of the full White test: With k regressors, there are $k + k + \binom{k}{2} = k(k+3)/2$ terms. For $k = 5$, that is 20 regressors in the auxiliary regression —

The White Test: Parsimonious Special Case

Insight: The fitted value $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_k x_{ik}$ is a linear combination of all x_j , so $\hat{y}_i^2$ is a linear combination of all cross-products.

White Test — Special Case (eq. 8.20)

Regress $\hat{u}_i^2$ on the OLS fitted values and their squares:

$$\hat{u}_i^2 = \delta_0 + \delta_1 \hat{y}_i + \delta_2 \hat{y}_i^2 + \text{error}$$

$H_0 : \delta_1 = \delta_2 = 0.$ $LM = nR_{\hat{u}^2}^2 \xrightarrow{d} \chi_2^2$ — only **2 degrees of freedom** regardless of k .

Advantage: Degrees of freedom do not grow with k . Powerful for detecting common heteroskedasticity patterns (variance increasing with $\hat{y}$).

Example 8.5: White Test on the Log Housing Price Equation

Recall: BP test on the log-log model failed to reject homoskedasticity ($p = 0.239$).

Special White Test

Regress $\hat{u}_i^2$ on $\widehat{\log(price)}_i$ and $\widehat{\log(price)}_i^2$:

$$R_{\hat{u}^2}^2 = 0.0392$$

$$LM = 88 \times 0.0392 = 3.45, \quad p\text{-value} = 0.178 \quad (2 \text{ d.f.})$$

- ▶ Fails to reject at 10% level (slightly more evidence than BP, but inconclusive).
- ▶ Both tests agree: the log-log model is approximately homoskedastic.

Weighted Least Squares (§8-4)

The Logic of WLS: Re-weighting Observations

Motivation: If we know how variance varies, we can do better than OLS.

WLS Setup (eq. 8.21)

Suppose the heteroskedasticity has the form:

$$\text{Var}(u_i | \mathbf{x}_i) = \sigma^2 h_i, \quad h_i = h(x_{i1}, \dots, x_{ik}) > 0$$

where h_i is known (or estimable).

WLS minimises the weighted sum of squared residuals (eq. 8.27):

$$\sum_{i=1}^n \frac{(y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_k x_{ik})^2}{h_i}$$

WLS as OLS on a Transformed Model

Divide the regression equation through by $\sqrt{h_i}$:

Starting from (eq. 8.24): $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i$

Dividing through gives the **transformed model** (eq. 8.25–8.26):

$$y_i^* = \beta_0 x_{i0}^* + \beta_1 x_{i1}^* + \cdots + \beta_k x_{ik}^* + u_i^*$$

where $y_i^* = y_i / \sqrt{h_i}$, $x_{ij}^* = x_{ij} / \sqrt{h_i}$, $x_{i0}^* = 1 / \sqrt{h_i}$, $u_i^* = u_i / \sqrt{h_i}$.

Proof that transformed model is homoskedastic:

$$\text{Var}(u_i^* \mid x_i) = \frac{\text{Var}(u_i \mid x_i)}{h_i} = \frac{\sigma^2 h_i}{h_i} = \sigma^2 \quad \checkmark$$

Since MLR.5 now holds for the transformed model, **OLS on the starred variables is BLUE** (Gauss-Markov).

WLS: Key Properties and Warning

If h_i is correctly specified:

- ▶ WLS is **unbiased** (linear transformation of the data)
- ▶ WLS is **BLUE** — lower variance than OLS
- ▶ WLS standard errors (from the transformed regression) are valid

Warning — misspecification risk:

If h_i is *wrong*, WLS is still consistent but:

- ▶ WLS standard errors may be **invalid**
- ▶ WLS may even be **less efficient** than OLS

Safe practice: After WLS, always compute *heteroskedasticity-robust SEs* for the WLS estimates. These are valid even if h_i is misspecified.

Example 8.6: Financial Wealth with WLS

Setup: Assume $\text{Var}(u_i \mid \text{inc}_i) = \sigma^2 \cdot \text{inc}_i$, so $h_i = \text{inc}_i$. Dataset: 401KSUBS.

Financial Wealth Equations (SEs in parentheses)

Variable	OLS	WLS ($w_i = 1/\text{inc}_i$)
<i>inc</i> (simple)	0.821 (0.104)	0.787 (0.063)
intercept	-10.57 (2.53)	-9.58 (1.65)
<i>inc</i> (expanded)	0.771 (0.100)	0.740 (0.064)
$(\text{age} - 25)^2$	0.0251 (0.0043)	0.0175 (0.0019)
<i>male</i>	2.48 (2.06)	1.84 (1.56)
<i>e401k</i>	6.89 (2.29)	5.19 (1.70)

Example 8.6: Interpretation — WLS vs. OLS

- ▶ WLS SEs are **uniformly smaller** — efficiency gain from correct weighting.
- ▶ 401(k) eligibility (*e401k*): coefficient drops from 6.89 (OLS) to 5.19 (WLS) — *economically meaningful* (roughly \$1,700 wealth difference).
- ▶ WLS relies on the assumption $\text{Var}(u \mid inc) \propto inc$. If wrong, efficiency advantage disappears.

Caution: If the variance function h_i is misspecified, always report **robust WLS SEs** rather than conventional WLS SEs to protect against invalid inference.

Feasible GLS: When h_i Must Be Estimated (§8-4b)

In practice, the variance function $h(x)$ is almost never known exactly.

FGLS: Exponential Variance Function (eq. 8.30)

Assume:

$$\text{Var}(u_i | \mathbf{x}_i) = \sigma^2 \exp(\delta_0 + \delta_1 \mathbf{x}_{i1} + \cdots + \delta_k \mathbf{x}_{ik})$$

Taking logs of $u_i^2 \approx \sigma^2 \exp(\delta_0 + \delta_1 \mathbf{x}_{i1} + \cdots + \delta_k \mathbf{x}_{ik}) \cdot \epsilon_i$:

$$\log(\hat{u}_i^2) = \alpha_0 + \delta_1 \mathbf{x}_{i1} + \cdots + \delta_k \mathbf{x}_{ik} + \text{error} \quad (\text{eq. 8.32})$$

Estimated weights: $\hat{h}_i = \exp(\hat{g}_i)$ where $\hat{g}_i$ are fitted values from [8.32].

Feasible GLS: Step-by-Step Algorithm

FGLS Procedure:

1. Estimate main equation by **OLS**; obtain residuals $\hat{u}_i$.
2. Compute $\log(\hat{u}_i^2)$.
3. Regress $\log(\hat{u}_i^2)$ on $x_{i1}, \dots, x_{ik}$; save fitted values $\hat{g}_i$.
4. Form estimated weights: $\hat{h}_i = \exp(\hat{g}_i)$.
5. Apply **WLS** with weights $w_i = 1/\hat{h}_i$.
6. Report **robust SEs** for the WLS step (in case h_i is misspecified).

Properties of FGLS:

- **Consistent and asymptotically more efficient** than OLS (if form is correct)

Example 8.7: Cigarette Demand — OLS vs. FGLS

Dataset: SMOKE ($n = 807$). BP test: $LM = 32.3$ ($p \approx 0$) — strong heteroskedasticity confirmed.

Cigarette Demand: OLS vs. FGLS (SEs in parentheses)

Variable	OLS	FGLS
$\log(\text{income})$	0.880 (0.728)	1.30 (0.44)
$\log(\text{cigprice})$	-0.751 (5.773)	-2.94 (4.46)
<i>educ</i>	-0.501 (0.167)	-0.463 (0.120)
<i>age</i>	0.771 (0.160)	0.482 (0.097)
<i>age</i> ²	-0.0090 (0.0017)	-0.0056 (0.0009)
<i>restaurn</i>	-2.83 (1.11)	-3.46 (0.80)
R^2	0.053	0.113

Example 8.7: Interpretation — FGLS Efficiency Gains

- ▶ FGLS SEs are **uniformly smaller** — substantial efficiency gain from re-weighting.
- ▶ Income elasticity: $0.880 \rightarrow 1.30$ (FGLS more precise and estimate shifts — OLS was noisy).
- ▶ Restaurant smoking restrictions: -3.46 cigs/day under FGLS vs. -2.83 OLS — larger and more precise.
- ▶ R^2 nearly doubles: $0.053 \rightarrow 0.113$ — FGLS fits the weighted data much better.

When heteroskedasticity is **strong** (BP $p \approx 0$), FGLS can dramatically improve efficiency and even shift point estimates. But remember: FGLS requires correct specification of $h(x)$.

Estimation Strategy Summary

Deciding which estimator and standard errors to use:

Situation	Estimator	Standard Errors
Homoskedasticity confirmed	OLS	Usual OLS SEs
Heterosk., form unknown	OLS	White robust SEs
Heterosk., form known	WLS	Usual WLS SEs
Heterosk., form estimated	FGLS	Robust WLS SEs

When in doubt: **OLS + robust SEs** is the defensible baseline. Pursue WLS/FGLS only when you have a strong theoretical or empirical reason to believe a specific variance function.

The Linear Probability Model Revisited (§8-5)

LPM Heteroskedasticity: The Theoretical Result

Recall from Lecture 6: the LPM has *built-in* heteroskedasticity.

Proof: For the LPM with $y_i \in \{0, 1\}$, given x_i , y_i is Bernoulli with mean $p_i = \mathbb{E}[y_i \mid x_i]$.

Therefore: $\text{Var}(y_i \mid x_i) = p_i(1 - p_i) = p(x_i)[1 - p(x_i)]$ (eq. 8.45)

Since $p(x_i) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik}$ depends on x_i , so does the variance. MLR.5 is **always violated** in the LPM.

Natural WLS weights for LPM (eq. 8.47):

$$\hat{h}_i = \hat{y}_i(1 - \hat{y}_i)$$

Constraint: WLS is only feasible if all $\hat{y}_i \in (0, 1)$. If any $\hat{y}_i \leq 0$ or $\hat{y}_i \geq 1$, weights are non-positive and WLS fails.

Example 8.8: Married Women's Labor Force Participation

Dataset: MROZ ($n = 753$). $y = inlf$ (1 = in labor force).

LPM for Labor Force Participation

Variable	Coef.	OLS SE	[Robust SE]
<i>nwifeinc</i>	-0.0034	(0.0014)	[0.0015]
<i>educ</i>	0.038	(0.007)	[0.007]
<i>exper</i>	0.039	(0.006)	[0.006]
<i>exper</i> ²	-0.00060	(0.00012)	[0.00012]
<i>age</i>	-0.016	(0.002)	[0.002]
<i>kidslt6</i>	-0.262	(0.034)	[0.034]
<i>kidsge6</i>	0.013	(0.013)	[0.013]
intercept	0.586	(0.154)	[0.151]
$n = 753, R^2 = 0.264$			

Example 8.9: PC Ownership — OLS vs. WLS for LPM

Dataset: GPA1 ($n = 141$). $y = PC$ (1 = owns computer).

PC Ownership: OLS vs. WLS

Variable	OLS	WLS ($w_i = 1/\hat{h}_i$)
<i>hsGPA</i>	0.065 (0.137)	0.033 (0.130)
<i>ACT</i>	0.00065 (0.0155)	0.0043 (0.0155)
<i>parcoll</i>	0.221 (0.093)	0.215 (0.086)
intercept	-0.0004 (0.490)	0.026 (0.477)
R^2	0.0415	0.0464

- ▶ OLS and WLS give very similar results — no efficiency concern here.
- ▶ *parcoll* is the only significant predictor: parental college degree

Summary and Decision Framework

Three-Step Decision Framework

Step 1 — Estimate by OLS; report with robust SEs.

OLS is unbiased and consistent under MLR.1–4. Robust SEs ensure valid inference regardless of σ_i^2 .

Step 2 — Test for heteroskedasticity (optional but informative).

Run BP test (χ_k^2) and/or White special case (χ_2^2). If rejected, proceed to Step 3.

Step 3 — Pursue efficiency if heteroskedasticity confirmed.

- ▶ *Known h_i* : use WLS with conventional WLS SEs.
- ▶ *Estimated h_i* : use FGLS with **robust** WLS SEs.
- ▶ *Unknown form*: stay with OLS + robust SEs.

Key Concepts Reviewed

The Problem: Heteroskedasticity ($\sigma_i^2 \neq \sigma^2$) leaves $\hat{\beta}_j$ unbiased but invalidates SEs and tests (MLR.5 violated).

Fix for Inference: White robust SE — $\widehat{\text{Var}}(\hat{\beta}_j) = \sum_i \hat{r}_{ij}^2 \hat{u}_i^2 / \text{SSR}_j^2$ — consistent under any σ_i^2 .

Testing: BP test ($\hat{u}^2$ on x_j, χ_k^2) or White special case ($\hat{u}^2$ on $\hat{y}, \hat{y}^2, \chi_2^2$).

Fix for Efficiency: WLS/FGLS achieves BLUE when $h_i = \text{Var}(u_i \mid x_i) / \sigma^2$ is correctly specified. Use robust WLS SEs when h_i is estimated.

Bridge to Lecture 8: Specification and Data Issues

Lecture 7 (today)

MLR.5 violated only:

- ▶ OLS is still unbiased and consistent
- ▶ Only SEs and tests are affected
- ▶ Fix: robust SEs or WLS

Lecture 8 (Chapter 9)

MLR.4 violated ($\mathbb{E}[u | x] \neq 0$):

- ▶ Omitted variable bias
- ▶ Functional form misspecification
- ▶ Measurement error (attenuation)
- ▶ Missing data / selection bias

Lecture	Assumption violated	Effect on OLS
7 (today)	MLR.5 only	Biased SEs: $\hat{\beta}$ consistent